

A gap in the proof of Lemma 3.12 of
Cavenagh–Greenhill–Wanless,
The cycle structure of two rows in a random Latin square

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Abstract

Case 3 of the splitting procedure used to prove Lemma 3.12 of [1] (cross-switch with respect to $\{\omega(j), \omega(j')\}$, then backflip with respect to $\{j, j'\}$) does not behave as claimed: the cross-switch reverses an ω -arc containing j' (or j), after which $\{j, j'\}$ is at ω -distance 2. When $\min(\alpha, \beta) \geq 3$ the backflip therefore produces a square of the wrong type. When $\min(\alpha, \beta) = 2 < \max(\alpha, \beta)$ the type is right, but the resulting edge is in general not an edge of the joining multigraph, so the identity $G_S = G_J$ on which the double count rests fails; already for $n = 5$, the sharp case, the splitting multigraph has vertices of degree 24 on the $S(\mu, F)$ side where the proof asserts at most 18. The lower bound of Lemma 3.12 survives (and is in fact elementary); the upper bound $3/2$, hence the upper bound of Theorem 3.13 and the results of Section 4 that use it (listed below), is not established by the published argument for any split other than $(2, 2)$. For $\min(\alpha, \beta) = 2$ the count can be corrected and gives the bound with 2 in place of $3/2$; for $\min(\alpha, \beta) \geq 3$ a canonical repair of case 3 exists when α is even and $\alpha < 2\beta$ (or symmetrically), and we do not see one when α, β are both odd. Exact data for $n \leq 11$ and sampled squares up to $n = 100$ indicate that the statement is true with much room to spare.

1 Setting and the procedure

We use the notation of [1], Section 3. L is a Latin square of order n containing a suitable partial square F ; ω is the permutation of the first m columns with $1 \circ \omega(j) = 2 \circ j$, whose cycle type $\lambda \in P(m)$ defines $S(\lambda, F)$. For columns j, j' let $\delta_{j, j'}$ be the row permutation with $\delta(i) \circ j = i \circ j'$; $\{j, j'\}$ is an *A-pair* if rows 1, 2 lie in different cycles of $\delta_{j, j'}$ and a *B-pair* otherwise. The *backflip* at an A-pair $\{j, j'\}$ lying on one ω -cycle of length $\alpha + \beta$ with $\omega^\alpha(j) = j'$ trades the $\delta_{j, j'}$ -cycle through row 2; it splits that ω -cycle into the two arcs between j and j' , of lengths α and β (Lemma 3.5). The *cross-switch* at a B-pair $\{j, j'\}$ with $\omega^\alpha(j) = j'$ (Definition 3.9) is a Latin bitrade (Q, Q') that preserves $S(\lambda, F)$ (Lemma 3.10).

Let μ be λ with one part $\alpha + \beta$ split into $\alpha, \beta \geq 2$. The splitting procedure (p. 294) considers, for $L \in S(\lambda, F)$, every pair $\{j, j'\}$ with $\omega^\alpha(j) = j'$ on an $(\alpha + \beta)$ -cycle and produces $L' \in S(\mu, F)$ as follows:

1. if $\{j, j'\}$ is an A-pair, backflip at $\{j, j'\}$;
2. else if $\{\omega j, \omega j'\}$ is an A-pair, backflip at $\{\omega j, \omega j'\}$;

3. otherwise cross-switch with respect to $\{\omega j, \omega j'\}$; “now $\{j, j'\}$ has become an A -pair, and we backflip with respect to $\{j, j'\}$ ”.

Item 4 asserts that in each case $L' \in S(\mu, F)$. Together with a fifth edge (a switch) this gives every $L \in S(\lambda, F)$ degree exactly $2(\mu_{\alpha+\beta} + 1)(\alpha + \beta)$ in a bipartite multigraph between $S(\lambda, F)$ and $S(\mu, F)$, while the joining procedure gives every $L \in S(\mu, F)$ degree between $\mu_\alpha \mu_\beta \alpha \beta$ and three times that; the two multigraphs are asserted to coincide, and the double count is Lemma 3.12:

$$\frac{1}{2} \leq \frac{|S(\lambda, F)| (\mu_{\alpha+\beta} + 1)(\alpha + \beta)}{|S(\mu, F)| \mu_\alpha \mu_\beta \alpha \beta} \leq \frac{3}{2} \quad (\alpha \neq \beta). \quad (1)$$

(For $\alpha = \beta$ the statement has $2(\mu_{2\alpha} + 1)/(\alpha \mu_\alpha (\mu_\alpha - 1))$ in place of the fraction; everything below applies verbatim, with the splits (α, α) , $\alpha \geq 3$, in the class $\min(\alpha, \beta) \geq 3$ and $(2, 2)$ the only unaffected split.)

2 The cross-switch reverses an arc

Write $e_1 = 1 \circ j$, $e_2 = 1 \circ j'$, $e_3 = 2 \circ j$, $e_4 = 2 \circ j'$ and consider the first form of Definition 3.9 ($\min(e_1, e_2, e_3, e_4) \in \{e_1, e_4\}$); the second form ($\min \in \{e_2, e_3\}$) is the first form applied to the ordered pair (j', j) , i.e. with the roles of j and j' (and of α and β) exchanged, so that the cross-switch at an unordered B -pair is well defined. On rows 1, 2 the bitrade is

$$Q' \supseteq \{(2, j, e_2), (1, j', e_3)\} \cup \{(1, \omega^k j, 2 \circ \omega^k j), (2, \omega^k j, 1 \circ \omega^k j) : 1 \leq k < \alpha\},$$

i.e. in every interior column of the ω -arc from j to j' the entries of rows 1, 2 are swapped, row 2 receives $e_2 = 1 \circ j'$ at column j , and row 1 receives $e_3 = 2 \circ j$ at column j' . (This is the definition; the proof of Lemma 3.10 confirms it: row 1 keeps the symbol set $\{1 \circ \omega^k j : 1 \leq k \leq \alpha\}$ on the cells $\{\omega^k j : 1 \leq k < \alpha\} \cup \{j'\}$, and “precisely one column from the pair $\{\omega^k j, \omega^{k'} j'\}$ has symbols swapped in rows 1 and 2”.) The remaining cells of Q lie in columns j, j' and in rows other than 1, 2; they do not affect ω .

Lemma 1. *Let L' be the cross-switch of L at the B -pair $\{j, j'\}$, $\omega^\alpha(j) = j'$, and let $\omega' = \omega_{L'}$. Then $\omega'(j) = \omega^{\alpha-1}(j)$, $\omega'(\omega^k j) = \omega^{k-1}(j)$ for $2 \leq k < \alpha$, $\omega'(\omega j) = j'$, and $\omega' = \omega$ on every other column. In words: the interior of the ω -arc from j to j' is traversed in reverse, and the cycle structure of ω is otherwise unchanged.*

Proof. Write $1', 2'$ for the new rows. Then $1' \circ j = e_1$, $1' \circ \omega^k j = 1 \circ \omega^{k+1} j$ for $1 \leq k < \alpha$ (since $2 \circ \omega^k j = 1 \circ \omega^{k+1} j$), $1' \circ j' = 1 \circ \omega j$; $2' \circ j = 1 \circ j' = 1 \circ \omega^\alpha j$, $2' \circ \omega^k j = 1 \circ \omega^k j$ for $1 \leq k < \alpha$, $2' \circ j' = e_4$. Now $\omega'(c)$ is the column whose row-1' entry equals $2' \circ c$: for $c = j$ this is $\omega^{\alpha-1} j$; for $c = \omega^k j$ with $2 \leq k < \alpha$ it is $\omega^{k-1} j$; for $c = \omega j$ it is j' (the symbol $1 \circ \omega j$ now sits at $(1', j')$); for $c = j'$ it is $\omega j'$, unchanged; for $c = \omega^{-1} j$, $2' \circ c = 2 \circ c = e_1 = 1' \circ j$, so $\omega'(c) = j$; all other columns are untouched in both rows. $\square$

The example on p. 293 of [1] shows exactly this: there $\omega = (1234567)$, $j = 2$, $j' = 5$, $\alpha = 3$, and the displayed cross-switch has rows 1245367 and 2534671, i.e. $\omega' = (1243567)$.

3 Case 3 lands in the wrong type

Proposition 2. *Let $\{j, j'\}$ and $\{J, J'\} = \{\omega j, \omega j'\}$ both be B -pairs with $\omega^\alpha j = j'$ on an $(\alpha + \beta)$ -cycle, and let L' be the cross-switch of L at $\{J, J'\}$ (either form). Then $\{j, j'\}$ is an A -pair of L' , but in the first form it is at ω' -distance 2 (from j to j'), and in the second form at ω' -distance 2 from j' to j . Consequently the backflip at $\{j, j'\}$ splits the $(\alpha + \beta)$ -cycle into parts 2 and $\alpha + \beta - 2$, which is the type μ only if $\min(\alpha, \beta) = 2$.*

Proof. First form. The arc from $J = \omega j$ to $J' = \omega^\alpha J$ has interior $\omega J, \dots, \omega^{\alpha-1} J$, and $\omega^{\alpha-1} J = \omega^\alpha j = j'$. By Lemma 1, $\omega'(j) = \omega'(\omega^{-1} J) = J$ and $\omega'(J) = \omega^{\alpha-1} J = j'$. The status of $\{j, j'\}$: column j' is an interior column of the arc, so rows 1, 2 are swapped there, while column j is untouched ($j \notin \{J, J'\}$ and j is not in the arc); hence $\delta_{j,j'}$ changes by a transposition of rows 1, 2 and $\{j, j'\}$ toggles from B to A , as claimed in [1]. The second form reverses the arc from J' to J , whose interior contains $j = \omega^{-1} J$ as its last column, and the same computation gives $\omega'(J') = j$, i.e. $\omega'^2(j') = j$. $\square$

So the asserted membership $L' \in S(\mu, F)$ in item 4 fails in case 3 whenever $\min(\alpha, \beta) \geq 3$.

The case $\min(\alpha, \beta) = 2$. Say $\alpha = 2 \leq \beta$. In the form of the cross-switch that reverses the arc of length 2 (a single interior column), nothing is reversed and case 3 is as described. In the other form the arc of length β is reversed; j or j' is its last interior column, the backflipped square has parts 2 and β , so it does lie in $S(\mu, F)$, and item 4 holds. But the proof of Lemma 3.12 rests on the identity $G_S = G_J$, and this fails: the case-3 edge $L-L'$ is supposed to be the joining rule-4 edge from L' (flip at $\{j, j'\}$ to L'' , then cross-switch at $\{\omega_{L''} j, \omega_{L''} j'\}$), which returns L by Corollary 3.11 only if $\{\omega_{L''} j, \omega_{L''} j'\} = \{J, J'\}$; and when the long arc has been reversed this is not so (e.g. in the second form $\omega_{L''}(j) = \omega^{\beta-2} J' \neq J$). Which form occurs is decided by the position of the smallest corner symbol, not by the author, so about half of the case-3 edges and their switch edges (item 5) are missing from G_J . We computed both multigraphs exactly for $n = 5$, $\lambda = (5)$, $\mu = (2, 3)$, $F = \emptyset$, over all 161280 Latin squares of order 5 (the procedures select trades by the smallest symbol, so one cannot normalise the first row): G_S and G_J each have 1036800 edges, but 172800 edges of G_S are not edges of G_J (exactly half of the case-3 edges and the switch edges of the same instances; all edges of cases 1 and 2 are in G_J), and the degrees of the μ -squares in G_S are 12, 18 and 24 (for 11520, 34560 and 11520 squares) against 18 for every μ -square in G_J . So the per-vertex assertion “each vertex of $S(\mu, F)$ has degree $d \leq 3\mu_\alpha\mu_\beta\alpha\beta$ ” is false for the splitting multigraph, though it holds on average over $S(\mu, F)$; the inequality $Z \leq 3\mu_\alpha\mu_\beta\alpha\beta|S(\mu, F)|$ itself therefore still holds at $n = 5$, with equality (the ratio is exactly 3/2 there), but not for the reason given. For $(4) \rightarrow (2, 2)$ at $n = 4$ the two multigraphs coincide, as they must: only arcs of length 2 occur. In summary, the published argument for the upper bound is incomplete for every split with $\max(\alpha, \beta) \geq 3$.

Numerical confirmation. We implemented Definition 3.9 literally (rows 1, 2 as above; in columns j, j' the rows $\delta^k(1)$, $1 \leq k < a$, with $\delta^a(1) = 2$, swap their two entries, exactly as in the definition — this is what balances the corner cells) and checked on uniformly sampled Latin squares (Jacobson–Matthews) that the cross-switch is a Latin square of the same type in every instance. Applying case 3 as written: in 1006 ordered instances (j, j') with $\min(\alpha, \beta) \geq 3$ at $n = 9$ (300 sampled squares) and 3098 at $n = 12$ (300 squares), the result had the wrong type in every single instance (e.g. $(2, 7)$ with $(\alpha, \beta) = (3, 4)$ gives $(2, 2, 5)$ instead of $(2, 3, 4)$; (9) with $(3, 6)$ gives $(2, 7)$).

4 What survives, and what depends on the gap

Only case 3 is affected. The joining rules 1–2 are valid and each rule-2 edge is a case-1 edge or the switch edge of a case-1 instance, so each $L \in S(\mu, F)$ has degree at least $\mu_\alpha \mu_\beta \alpha \beta$ in the multigraph of valid splitting edges, in which each $L \in S(\lambda, F)$ has degree at most $2(\mu_{\alpha+\beta} + 1)(\alpha + \beta)$. Hence the *lower* bound $\frac{1}{2}$ in (1) stands. (In fact it is elementary: with X the set of pairs $(L, \{j, j'\})$, $L \in S(\lambda, F)$, $\omega^\alpha j = j'$, and J the analogous set of joining pairs, the flip and the switch give $|J| = 2|X_A|$ exactly, so (1) reads $\frac{1}{3} \leq |X_A|/|X| \leq 1$, and the lower bound $\frac{1}{2}$ is $|X| \geq |X_A|$.) The *upper* bound $\frac{3}{2}$, equivalently

$$\Pr[\{j, j'\} \text{ is an } A\text{-pair}] \geq \frac{1}{3} \quad \text{for } (L, \{j, j'\}) \text{ uniform on } X,$$

is not established by the published argument for $\max(\alpha, \beta) \geq 3$; Proposition 3 below establishes it with $\frac{1}{4}$ in place of $\frac{1}{3}$ for $\min(\alpha, \beta) = 2$, so it remains open for $\min(\alpha, \beta) \geq 3$.

Within [1] the upper bound of Lemma 3.12 is used for: the upper bound of Theorem 3.13; the lower bound $(2/3)^{\kappa-1}$ of Lemma 4.1 and, in Lemma 4.2, the upper bound whenever the path from λ to μ contains a splitting step and the lower bound whenever it contains a joining step; the bounds on $\mathbb{P}_F(E)$ of Theorem 4.3 (whose proof invokes the equality of the two multigraphs “as in the proof of Lemma 3.12”); the upper bound $\mathbb{P}_F((m)) \leq 2m^{-2/3}$ of Lemma 4.4; both halves of Corollary 4.5 (its lower half uses the $(2/3)^{\kappa-1}$ bound, its upper half the factor $m^{1/3}$ from Lemma 4.4); and the results that rest on these: Corollary 4.6, Lemma 4.7, Theorem 4.9, Corollary 4.10 and Theorem 4.11. (Corollary 4.10, on the number of intercalates, is in any case superseded by Kwan and Sudakov [4].) The upper bound $2^{\kappa-1}$ of Lemma 4.1, the lower bound $\mathbb{P}_F((m)) \geq 2m^{-2}$ of Lemma 4.4, Corollary 4.8, and the upper bounds on short-cycle counts derived from the joining direction are unaffected. For instance, from the joining direction alone one still gets $\mathbb{E}[\ell N_\ell] \leq 2m/(m - \ell)$ for the number N_ℓ of ℓ -cycles, $\ell \leq m/2$ (sum the inequality $\mathbb{E}[\ell N_\ell \beta N_\beta] \leq 2\mathbb{E}[(\ell + \beta)N_{\ell+\beta}]$, with $N_\ell N_\beta$ read as $N_\ell(N_\ell - 1)$ when $\beta = \ell$, over β), and by joining k cycles successively one obtains a factorial-moment bound of the form $\mathbb{E}[(\kappa)_k] \leq 2^{k-1}(4 + k^2/\log m)(\log m)^k$ for the number of cycles κ , valid for every k (we give the derivation elsewhere). With $k = \log m$ this gives $\Pr[\kappa \geq 16 \log m] = O(m^{-2} \log m)$, and with $k = \lceil \sqrt{m} \rceil$ it gives, by Markov’s inequality on $(\kappa)_k \geq (8\sqrt{m})^k$,

$$\Pr[\kappa \geq 9\sqrt{m}] \leq m \left(\frac{\log m}{4\sqrt{m}} \right)^{\sqrt{m}} = \exp\left(-\left(\frac{1}{2} - o(1)\right)\sqrt{m} \log m\right),$$

which is the conclusion of Theorem 4.9 with a better rate. So the statements of Theorem 4.9 and of the $\log m$ -scale tail survive (by a different proof), while the lower bound on the number of cycles of Theorem 4.11 does not. Likewise the upper half of Corollary 4.5 survives in a weaker form: joining the κ parts of a type one pair at a time and using at each step the surviving inequality $\mathcal{C}(\text{finer}) \leq 2\mathcal{C}(\text{coarser})$ (valid also when the two joined parts are equal) gives $\mathcal{C}(\lambda) \leq 2^{\kappa-1}\mathcal{C}((m))$ for a type λ with κ parts, hence

$$\mathbb{P}_F(\lambda) \leq \frac{\gamma(\lambda) 2^{\kappa-1} \mathcal{C}((m))}{\gamma((m)) \mathcal{C}((m))} \leq \frac{m}{2} \prod_{i=1}^{\kappa} \frac{2}{c_i}$$

for a type with parts $c_1, \dots, c_\kappa$, where $\gamma(\lambda) \leq m!/\prod c_i$ is the number of permutations of type λ , $\gamma((m)) = (m-1)!$, and $\mathbb{P}_F(\lambda) \propto \gamma(\lambda)\mathcal{C}(\lambda)$ (in place of the factor $m^{1/3}$, which came from Lemma 4.4).

A downstream use. Gill, Mammoliti and Wanless [3] use Corollary 4.5 (upper half) and Theorem 4.9 to prove their Theorem 9 (with probability $1 - o(\exp(-\frac{1}{4}n^{1/2}))$) the first two rows of a random Latin square contain a row cycle of length at least $n^{1/2} \log^{1-\varepsilon} n$, on which the average-case bound of their Theorem 1 rests through Theorem 12. Both inputs are affected by the gap, but both are replaced by the surviving bounds just stated: with $\mathbb{P}_F \leq \frac{m}{2} \prod (2/c_i)$ in place of $m^{1/3} \prod (2/c_i)$ the estimate in their proof becomes $\exp(-\delta\sqrt{n}(1-o(1)) \log^\varepsilon n)$ times a polynomial factor, and the number of cycles is at most $9\sqrt{n}$ with probability $1 - \exp(-\Omega(\sqrt{n} \log n))$; so their Theorem 9 holds as stated.

Cavenagh and Wanless [2] use three results of [1] in the proof of their Theorem 1.1 (asymptotically the same number of Latin squares of each parity): the upper half of Corollary 4.5 in their Lemma 2.1 (no row cycle of length $\geq n - \log n$ in the last two rows, with probability $1 - o(1)$), Theorem 4.9 in their Lemma 2.2, and the upper bound of Theorem 3.13 in their Lemma 2.3, for the splits of an even part z into two odd parts $a, z - a$ (so $\min(a, z - a) \geq 3$). The second use is repaired as above (our Proposition is stated for $F = \emptyset$; its proof uses only flips and switches at marks in the free columns and trades along cycles outside F , so it applies in $\Omega(F)$ with m in place of n , but we have not written this out). The first is not repaired by the weaker surviving constant: with $n/2$ in place of $n^{1/3}$ the estimate in Lemma 2.1 becomes $O(2^{(\log n)/2} \log n)$, and already its $\lambda = (n)$ term is 1; what Lemma 2.1 needs is that the cycle through column 1 has length $n - \omega(1)$ with probability $1 - o(1)$, which includes $\mathbb{P}_F((n)) = o(1)$, discussed below. The third is the unproved direction at exactly the splits where case 3 fails, outside the range of Proposition 3; what their Lemma 2.3 actually needs is weaker than $\frac{3}{2}$ — any constant, or even the averaged statement $\sum_{\mu \in M} \gamma(\mu) |S(\mu, F)| / (\gamma(\lambda) |S(\lambda, F)|) \rightarrow \infty$ over the odd splits M of a large even part — but we do not have that either. So, as far as we can see, Theorem 1.1 of [2] currently depends on the gap at two points.

More generally, the surviving direction bounds fine types from above only: it gives no upper bound on the probability of a coarse type. In particular we know of no proof, from the surviving material or from our own identities, that the first two rows of a random Latin square form a single n -cycle with probability $o(1)$ (the statement $\mathbb{P}_F((m)) \leq 2m^{-2/3}$ of Lemma 4.4). What is available is a constant: by row-pair symmetry and the intercalate count of [4], $\mathbb{E}N_2 \geq \frac{1}{2} - o(1)$; the $\ell = \beta = 2$ case of the surviving inequality gives $\mathbb{E}[N_2(N_2 - 1)] \leq 2\mathbb{E}N_4 \leq m/(m - 4)$; so by the second moment method $\Pr[N_2 \geq 1] \geq (\mathbb{E}N_2)^2 / \mathbb{E}[N_2^2] \geq \frac{1}{6} - o(1)$ and $\mathbb{P}_F((m)) \leq \frac{5}{6} + o(1)$. Proposition 3 does not improve this: it gives $\mathbb{P}_F((m)) \leq 4\mathbb{P}_F((2, m - 2))$, the surviving inequality gives $\mathbb{P}_F((2, m - 2)) \leq \frac{m}{2(m - 2)} \mathbb{P}_F((m))$, and iterating over k two-cycles loses a factor of order $4^k k!$. (We checked the other papers citing [1] that are indexed by MathSciNet; the remaining citations are to the intercalate estimates, superseded by [4], or to the conjecture on multiplication groups of random loops.)

5 A partial repair

A cross-switch at a B -pair $\{J, J'\}$ at any ω -distance $a \geq 2$ is a type-preserving trade (the proof of Lemma 3.10 does not use $a = \alpha$), and by Lemma 1 it reverses the arc from J to J' . If $\{J, J'\}$ is disjoint from $\{j, j'\}$, it toggles the status of $\{j, j'\}$ iff exactly one of j, j' is an interior column of that arc, say at position i ($1 \leq i < a$), and then the ω -distance from j to j' changes by $\pm(a - 2i)$ (sign according to which of j, j' is interior). Thus, among cross-switches at pairs disjoint from $\{j, j'\}$, the status-toggling moves that preserve the distance are precisely those at B -pairs *symmetric about j or about j'* ($a = 2i$).

Suppose α is even and $\alpha < 2\beta$, and let $R = \{\omega^{\alpha/2}j, \omega^{3\alpha/2}j\}$: the pair at distance α symmetric about j' , whose arc avoids j . If R is a B -pair, cross-switch at R : $\{j, j'\}$ becomes an A -pair whose two ω' -arcs still have lengths α and β (in the form that reverses the arc of R through j' the distance from j to j' stays α ; in the other form it becomes β), and the backflip at $\{j, j'\}$ lies in $S(\mu, F)$. If R is an A -pair, backflip at R itself, which is a pair at distance α . Either way case 3 acquires a valid edge (verified on 3928 sampled instances at $n = 10, 13$: type μ in all of them). The same works with the roles of α, β exchanged when β is even and $\beta < 2\alpha$. The multiplicities on the $S(\mu, F)$ side change (an A -pair's backflip may now serve two splitting pairs), so the constant $3/2$ would have to be recomputed.

When α and β are both odd, every distance-preserving toggle has even arc length, so no symmetric pair is at distance α or β ; if all symmetric pairs about j and j' happen to be A -pairs, no single cross-switch repairs $\{j, j'\}$, and two-step repairs pass through a third type, which breaks the two-sided count. We do not see a canonical repair in this case (nor when $\alpha \geq 2\beta$ with β odd). Repairs that use other rows do exist — for a B -pair $\{j, j'\}$ with column cycle $Z \ni 1, 2$, trade rows z, w lying on different arcs of Z between 1 and 2 along the row cycle of $\{z, w\}$ through j (if j, j' lie in different row cycles of $\{z, w\}$; this separates 1 from 2 and keeps the type), or cross-switch rows z, w at $\{j, j'\}$ (if they lie in one; this changes the type to μ) — but they are indexed by a free pair of rows and so have unbounded multiplicity on the $S(\mu, F)$ side, which defeats the double count.

6 The case $\min(\alpha, \beta) = 2$: the bound holds with constant 2

For $\min(\alpha, \beta) = 2$ the published count can be corrected at the price of a weaker constant. Say $\alpha = 2 \leq \beta$, $m = \beta + 2$.

Proposition 3. *If $\alpha = 2$ (or $\beta = 2$), then*

$$\frac{|S(\lambda, F)| (\mu_m + 1) m}{|S(\mu, F)| \mu_2 \mu_\beta 2\beta} \leq 2 \quad (\beta \neq 2),$$

and for $\alpha = \beta = 2$ the quantity of Lemma 3.12 is ≤ 2 as well.

Proof. Every splitting instance $(L, \{j, j'\})$ yields two squares in $S(\mu, F)$: in cases 1 and 2 this is item 4, and in case 3 the cross-switch either reverses nothing (the arc of length 2) or reverses the arc of length β , and in both forms the backflip at $\{j, j'\}$ produces parts 2 and β (Proposition 2). So $Z = 2(\mu_m + 1)m |S(\lambda, F)|$ counts valid edges. It remains to show that every $M \in S(\mu, F)$ receives at most 4 edges per joining pair, i.e. per pair $\{q, q'\}$ with q in a 2-cycle and q' in a β -cycle of ω_M . Attribute each edge (L, M) to a joining pair of M as follows: the direct edge of case $c \in \{1, 2, 3\}$ to the pair at which the backflip was made ($\{j, j'\}$ in cases 1, 3; $\{\omega j, \omega j'\}$ in case 2), which is an A -pair of M (the backflipped pair is an A -pair afterwards); the switch edge of cases 1 and 3 to the switched pair $\{j, j'\}$, which is a B -pair of M (Lemma 3.7: it was an A -pair before the switch); and the switch edge of case 2 to $\{\omega j, \omega j'\}$, which is a B -pair of M (it was the backflipped A -pair of L' , and the switch at $\{j, j'\}$ trades the row cycle through j or through j' , which contains exactly one of $\omega j, \omega j'$, so the status toggles by the argument in the proof of Lemma 3.7). Now fix M and a joining pair Q of M and count the edges attributed to Q .

If Q is an A -pair of M , only direct edges arrive, and $L_1 := \text{flip}_Q(M)$ is determined. Case 1 contributes (L_1, Q) ; case 2 contributes $(L_1, \{\omega_{L_1}^{-1}q, \omega_{L_1}^{-1}q'\})$; in case 3, L is a cross-switch of L_1

at a B -pair $\{J, J'\}$ with $Q = \{j, j'\}$ at ω_L -distance 2, and $\{J, J'\}$ is determined by L_1 and Q : let (a, b) be the ordering of Q with $\omega_{L_1}^2(a) = b$ (unique since $\beta \geq 3$; for $\beta = 2$ both orderings give the same candidate below and there is no long arc, so case 3 then contributes at most 1); if nothing was reversed, $\{J, J'\} = \{\omega_{L_1}a, \omega_{L_1}b\}$; if the long arc was reversed, the computation of Proposition 2 (with j' the column of Q at distance 2 after j in L) gives $\omega_{L_1}(j') = J'$ and $\omega_{L_1}^{-1}(j') = J$, i.e. $(a, b) = (j', j)$ and $\{J, J'\} = \{\omega_{L_1}^{-1}a, \omega_{L_1}a\}$, the pair symmetric about a . So case 3 contributes at most 2 edges. Total ≤ 4 .

If Q is a B -pair of M , only switch edges arrive. For cases 1 and 3 the switch was made at Q itself, $M' = \text{sw}_Q(M)$ is determined (the switch is an involution: it preserves the symbol sets of the two row cycles, so the smallest-symbol rule re-selects the same cycle), and the direct edge into M' at the A -pair Q is the case-1 edge or one of the two case-3 edges listed above: ≤ 3 edges. For case 2, $Q = \{\omega j, \omega j'\}$ and the switch was made at $\{j, j'\}$; the switch inverted the row cycle of M' through j or through j' (whichever carries the smaller symbol, which is read off from M since the symbol sets are preserved), so $j = \omega_M(\omega j)$ if the cycle through j was inverted and $j = \omega_M^{-1}(\omega j)$ otherwise, and similarly j' ; then $M' = \text{sw}_{\{j, j'\}}(M)$ and $L = \text{flip}_Q(M')$ are determined: 1 edge. Total ≤ 4 .

Hence $Z \leq 4\mu_2\mu_\beta 2\beta |S(\mu, F)|$ (resp. $4\binom{\mu_2}{2}4|S(\mu, F)|$ for $\alpha = \beta = 2$), and the result follows as in the proof of Lemma 3.12. $\square$

We verified the recipe by computer: on sampled squares at $n = 7, 10, 12$ every direct and every switch edge of every splitting instance with $\alpha = 2$ is recovered from its target and attributed pair by the inverse recipes above, with at most 4 candidates per pair (`s13_cgw_min2.py`); and in the exact $n = 5$ computation of Section 3 the multiplicities per (target, attributed pair) are exactly 2, 3 or 4, direct edges always arrive at A -pairs and switch edges at B -pairs. The constant 2 is what this count gives at $n = 5$ ($24 = 4 \cdot 6$ is attained), where the truth is $3/2$.

In the language of Section 4, Proposition 3 says that for a marked pair at distance 2 (or $m-2$) on an m -cycle, $\Pr_X[A] \geq \frac{1}{4}$: a lower bound on a different-cycles event, which the surviving direction alone could not give. The reason it works for $\alpha = 2$ and not for $\min(\alpha, \beta) \geq 3$ is not multiplicity but type: for $\min(\alpha, \beta) \geq 3$ the wrong-form case-3 edges land in $S(\mu', F)$ with μ' the $(2, m-2)$ split, so the same count gives only $2(|X| - |X_3|) \leq 3|J|$, where X_3 is the set of instances in case 3 ($\{j, j'\}$ and $\{\omega j, \omega j'\}$ both B -pairs), i.e. $\Pr_X[A] \geq \frac{1}{3}(1 - \Pr_X[\text{case 3}])$, which is empty without a lower bound on $\Pr_X[A]$ at $\{j, j'\}$ or at $\{\omega j, \omega j'\}$.

7 What the data say

The statement of Lemma 3.12 is almost certainly true, and far from sharp for $n \geq 6$. By the identity $|J| = 2|X_A|$ the quantity in (1) is $\mathcal{C}_n(\lambda)/\mathcal{C}_n(\mu)$, where $\mathcal{C}_n(\lambda)$ is the number of completions of two rows of type λ , and $\Pr_X[A] = \mathcal{C}_n(\mu)/(2\mathcal{C}_n(\lambda))$. From the exact values of $\mathcal{C}_n(\lambda)$ tabulated in [1] (Tables 1 and 2, $n \leq 11$):

n	#splits $(\lambda \rightarrow \mu)$	$\max \mathcal{C}_n(\lambda)/\mathcal{C}_n(\mu)$	$\min \Pr_X[A]$
5	1	1.5000 (sharp)	0.3333
6	3	0.9545	0.5238
7	4	1.0035	0.4982
8	8	1.0017	0.4991
9	11	1.0002	0.4999
10	19	1.0000	0.5000
11	26	1.0000	0.5000

(all splits with $\alpha, \beta \geq 2$; for $n \geq 7$ every ratio lies in $[0.985, 1.0035]$, with the minima 0.9861 at $(3, 4) \rightarrow (2, 2, 3)$, $n = 7$, and 0.9853 at $(2, 2, 4) \rightarrow (2, 2, 2, 2)$, $n = 8$, and the excess above 1 at most 0.35%, in the covered and the uncovered classes alike — e.g. $(2, 6) \rightarrow (2, 3, 3)$ at $n = 8$, in the $(3, 3)$ class, has ratio 1.0017). On uniformly sampled squares with a uniformly random marked pair at distance $\alpha \geq 2$ on an m -cycle, $m - \alpha \geq 2$, we find $\Pr_X[B] - \Pr_X[A] = -0.022 \pm 0.011$, $+0.008 \pm 0.012$, -0.030 ± 0.018 at $n = 30, 50, 100$ (9000, 7200, 3600 instances), i.e. $\Pr_X[A] = 0.50 \pm 0.01$ throughout. So the true constant in the upper bound of (1) appears to be $1 + o(1)$ (the maximal excess over 1 in the table is $3.5 \cdot 10^{-3}$, $1.7 \cdot 10^{-3}$, $1.8 \cdot 10^{-4}$, $< 10^{-5}$, $< 10^{-5}$ for $n = 7, \dots, 11$) rather than $3/2$; the question is only what has been proved.

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